SPEACS-2: Intensive Care Unit "Communication Rounds" with Speech Language Pathology

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Intensive care unit (ICU) nurses occupy an essential role in facilitating patient communication and preventing the detrimental effects experienced by critically ill patients who are unable to speak, yet most are not equipped with the tools or training to enable communication most effectively with patients who are unable to speak. The goal of the Study of Patient-Nurse Effectiveness with Assisted Communication Strategies (SPEACS-2) is to explore the impact of an innovative, Web-based instructional package for ICU nurses with pocket reference guides, an instructional manual, and the provision of "low-tech" augmentative and alternative communication materials on nursing care quality and patient clinical outcomes. We hypothesize that this intervention will 1) improve nurses' skills in assessing and communicating with ICU patients who are unable to speak and 2) increase the collaboration between nursing and speech-language pathology in addressing complex patient communication needs in the ICU. (*Geriatr Nurs* 2010;31:170-177)

Critically ill older adults are at greater risk for impaired communication with family members, nurses, and other care providers than their younger adult counterparts because of preexisting communication disabilities, such as poor vision, poor hearing, slowed response time, dementia, and/or language impairments (e.g., aphasia),[1-3] as well as their higher risk of experiencing delirium during critical illness.[4] Respiratory tract intubation imposes additional communication disability—loss of voice—on many of the several million older adults treated in the intensive care unit (ICU) each year.[5-8] The Joint Commission on Accreditation of Healthcare Organizations recently released approved standards that will require identification of the patient's oral and written communication needs, including "the need for personal devices such as hearing aids or glasses, language interpreters, communication boards, and translated or plain language materials" (*www.patientprovidercommunication.org/pdf/8.pdf*).

ICU nurses occupy an essential role in facilitating patient communication and preventing the detrimental effects experienced by critically ill patients who are unable to speak, yet most are not equipped with the tools or training to most effectively enable communication with patients who are unable to speak. Nurses learn how to communicate with nonspeaking ICU patients by trial and error and admit to feeling frustrated and avoiding nonspeaking patients whose nonvocal messages are difficult to understand.[9-11] Lack of appropriate training of communication materials and access to communication experts, such as speech language pathologists, prevent ICU nurses from adequately addressing patients' communication needs.[9,12]

To address these deficiencies, we have undertaken a Robert Wood Johnson Foundation Interdisciplinary Nursing Quality Research Initiative funded project, the *SPEACS-2: Improving Patient Communication and Quality Outcomes in the ICU*. The goal of SPEACS-2 is to explore the impact of an innovative, Web-based instructional package for ICU nurses with pocket reference guides, an instructional manual, and the provision of "low-tech" augmentative and alternative communication (AAC) materials on nursing care quality and patient clinical outcomes. We hypothesize that this intervention will 1) improve nurses' skills in assessing and communicating with ICU patients who are unable to speak and 2) increase the collaboration between nursing and speech-language pathology in addressing complex patient communication needs in the ICU. In this article, we describe weekly communication case conferences led by a speech-language pathologist (SLP; author BB) at the University of Pittsburgh Medical Center as part of the SPEACS-2 intervention. These cases highlight communication problems typically experienced by older adults in acute and critical care settings and illustrate a novel approach to practice change.

Methods

The SPEACS-2 Communication Skills Training program is divided into 6 self-learning modules using video exemplars of target skills. Modules are organized according to a communication assessment algorithm to reinforce assessment and decision-making steps in choosing and applying AAC strategies or requesting speech language consultation with ICU patients. Nurses receive 1.5 continuing education credits for program completion. Selected communication resource nurses on each unit receive more in-depth training to serve as peer mentors and champions for the program. Teaching posters highlighting the "Communication Strategy of the Week" are displayed on the study units each week during the 3-month training period.

Communication materials are supplied to the unit in a SPEACS-2 "communication cart." The SPEACS-2 algorithm and intervention materials were previously developed and tested in the Study of Patient-Nurse Effectiveness with Assisted Communication Strategies (SPEACS) (5R01-HD042988) (M.B. Happ, K.L. Garrett et al, 2003-2008). The SLP leads a weekly communication case conference to illustrate and reinforce communication assessment and assistive strategies with existing patients. SLPs are experts in communication disorders and AAC systems. AAC consultation services provided by SLPs to acute and critical care units have been previously described[13-19] but are not yet standard care in ICUs nationwide. Importantly, SLPs can conduct a thorough assessment of a patient's communication ability and match communication strategies and materials to the individual's abilities, needs, and preferences. Weekly case conferences are documented in structured field notes by the research coordinator (J.S.). The following communication case conferences conducted by the SLP (B.B.) with patients in the Neurological ICU and Trauma ICU illustrate assessment and communication strategies that move beyond the basic first-line approaches commonly used by ICU nurses when working with critically ill older adults. Pseudonyms are used for patients.

Case 1. Mark Manley

The patient, Mr. Manley, was in the early, acute stages of recovery from stroke in the Neurological ICU. He was alert and oriented to person and place but not to date or time. He received oxygen via a tracheostomy. His vision was poor even with glasses. When the night-shift nurse offered Mr. Manley an alphabet board and felt-tip pen with a clipboard, she found that the patient's vision was too poor to see the alphabet board or to write legibly. This discovery may have qualified him as legally blind as a result of the stroke. The patient could nod or shake his head consistently and appropriately in response to yes-no questions. He could mouth words, gesture with both hands, point to areas of his body, give thumbs up and down, and hold up fingers with both hands accurately on command (e.g., "hold up two fingers").

Mr. Manley's nurse communicated with the patient by using yes-no questions and gestures. After receiving the night nurse's information about the patient's vision impairment, she did not try using any other communication tools from the communication cart.

The SLP completed a brief assessment during which the patient was unable to read an enlarged alphabet/picture board. The SLP recommended the following AAC strategies:

- Instruct the patient to mouth the first letter or use partner (nurse)-assisted pointing to the first letter on the enlarged alphabet board, then mouth the whole word.[20,21]
- Ask very specific questions or use partner (nurse)-assisted pointing to topic category to get short (mouthed) answers from patient.[22]
- Teach patient specific gestures for messages such as "I have a dry mouth" or "I need to be suctioned."[23]

SPEACS-2 COMMUNICATION CARE PLAN

1. This patient likes to talk about:

Ice Chips Temperature

Family Sports

2. Please make sure that patient ALWAYS has:

Glasses Notebook and Felt Tip Pen

Category Boards Large Alphabet Board

3. Communication Tips:

Eye Contact/Engage Exaggerate Mouthing Words

Ask SPECIFIC Questions Dramatic Gestures

4. Communication Strategies that work BEST:

First-Letter Spelling with Mouthing Words

Face Expressions Gestures Pick Topic Category

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Figure 1. Communication Care Plan.

- Encourage patient to exaggerate mouthed letters and words, use facial expressions, use arms and hands in gestures, and add drama to gestures for increased effect.
- Provide praise and positive reinforcement when the patient uses effective gestures.
- Use eye contact to engage the patient.
- Remind him to be dramatic in his gestures.

The SLP completed a Communication Care Plan Wall Card (see Figure 1) and recommended a full SLP consult to 1) help with strategies and 2) provide very large font communication boards and other tools that are not typically available on the unit. The SLP service received a consultation request for tracheostomy speaking valve; however, the tracheostomy required downsizing before a speaking valve could be implemented.

Case 2. Roy Patterson

Mr. Patterson, a patient in the Neurological ICU, was ventilated by tracheostomy for several days following brain surgery. He was arousable to voice at normal volume, oriented to person and place, and able to follow 1-step commands consistently. He was able to nod or shake his head very slightly in response to yes-no questions. Mr. Patterson's nurse attempted to use head nods and shakes as a yes-no signal with limited success. After a few successful head-nod responses at the beginning of encounters followed by no response from the patient, the nurse surmised that the patient was no longer interested in communicating.

The SLP provided a more detailed assessment of the patient's ability to signal yes-no, discovering that Mr. Patterson became quickly fatigued using the head nod or shake motions, tiring to the point of being unable to move his head at all after 1 or 2 questions. He had right-sided hemiparesis and was able to move his left hand only slightly; he held up 1 or 2 fingers and signaled by pointing up or down but tired quickly. He was only able to use his hand in this way to respond to 3 or 4 questions before he could no longer raise his hand. Mr. Patterson was able to look up or look down with eye movements to signal yes or no, responding consistently and appropriately to questions with this signal. He was able to open his mouth slightly (a few centimeters) and stick his tongue out just past his teeth but could not purse his lips or smile. He tired easily with this motion.

The SLP finalized a yes-no signal for Mr. Patterson that he could consistently use to answer questions. Tagging the end of the yes-no question provides focus for patients who may be cognitively impaired, sedated, or delirious and provides them with a visual reminder to use the yes-no signal.[24] For Mr. Patterson, the SLP recommended using upward-downward eye gaze for yes-no. The "tag" and signal at the end of the question is performed as "Yes?"(voice inflection upward with hand waved horizontally above the patient's forehead) [pause] "or no? (voice drops down and hand is waved below the patient's chin). Using this technique, the patient correctly identified his name and location and clearly signaled "yes" to the question "Do you like football?" However, he was unable to recognize the day and year.

In summary, the SLP recommended the following AAC strategies:

- Establish eye contact.
- Use tagged yes-no questions with a signal.[24]
- Establish individual signals, for example, sticking out tongue for suctioning or pointing up to signal a need to communicate.[17]
- As the patient's energy and attention improve, use the yes-no signal to indicate choices with partner (nurse)-assisted pointing to topic categories, messages on a picture communication board, or alphabet letters.[22]

Although the ICU nurses were preparing to transfer Mr. Patterson to a rehabilitation facility the following day, a Yes-No Signal Wall Card was made and posted so that all caregivers and visitors could be consistent in the use and interpretation of the upward-downward eye gaze signal. The nurses acknowledged that this was important individualized information to share with rehabilitation staff during transfer of care and agreed that the Yes-No Signal Wall Card should travel with the patient to the new facility.

Case 3. Patricia O'Connor

Mrs. O'Connor was a 92-year-old patient in the Trauma ICU who sustained a fall with injury and was receiving mechanical ventilation via tracheostomy. She was profoundly hard of hearing and wore a working hearing aid in her right ear. The left hearing aid was broken. She had very poor eye sight and wore glasses but was unable to see well enough to read (large) letter cards that her family made available for her at the bedside. Mrs. O'Connor nodded consistently and appropriately to yes-no questions. She mouthed words but was difficult to understand. She could point but most often refused to do so and did not have the strength or dexterity to write.

Mrs. O'Connor's nurse had little success using the large picture communication cards with the patient. The Audiology Service was consulted at the request of the communication resource nurse for a head phone hearing amplification device. This arrived just before the case conference, which helped considerably when the SLP spoke directly into the device microphone for the patient to hear her. The SLP encouraged the patient to exaggerate her articulation and mouth words slowly, which the patient learned quickly and tried diligently to use. The patient was previously described as "angry" and "depressed"; however, this method captured the patient's attention. Each success engendered more enthusiasm and motivation until the patient finally became too tired to continue. The SLP primarily wrote key words in large print to ask questions (e.g., "Where are you from?").[25] Written multiple choices worked well for some of the patient's responses. Choices were written in large print and announced aloud (Figure 2).[26,27] The patient mouthed her answer rather than pointing to a choice on the paper.

The patient's family attended the conference, expressing appreciation and an earnest desire to communicate better with Mrs. O'Connor. The SLP recommended that the nurse and family keep written message choices, keywords, and pictures in the notebook for future reference. She instructed the family to establish gestures and favorite topics with the patient. The family was encouraged to add this individualized information about the patient to the Communication Care Plan Wall Card started by the SLP. Nurses, previously frustrated at the communication impasse with this patient, expressed gratitude and relief. "This is the perfect patient to get your help, it has been so frustrating trying to communicate with her! Thank you!" A full SLP consultation was recommended for reasons similar to those in Mr. Manley's case, described earlier.

Figure 2. Written choice communication strategy.[26,27] Choices for geographic place can also be arranged on a map or in a compass format.

The SLP recommended the following AAC strategies:

- Use the amplification device Listenator, speaking directly into the microphone.
- Patient should wear her glasses.
- Encourage patient to overarticulate when mouthing words, repeatedly if necessary.
- Remind patient to slow down when mouthing.
- Use mouthing with first-letter spelling:
    - Because the patient has poor vision, instruct her to *mouth* the first letter of the desired word before mouthing the whole word.
- Use large picture boards.
- Use procedure explanation boards.
- Use written choice responses in large print.
- Write key words in large print to increase patient comprehension.
- Enlist the family's help to establish gestures, common messages, and topics the patient enjoys.

Discussion

Interdisciplinary rounds and bedside case presentations are not the norm for teaching/learning and nursing staff development. In one study, nurses assumed passive roles during interdisciplinary rounds, likely because of the power structure between nurses and physicians.[28] The collaborative and interactive structure of our bedside case conference is preferable to conference-room presentations for patient education and nursing staff development.[29,30] This is an example of "situated learning"—the acquisition of skills, practice, and culture within the setting and situation in which practice normally occurs.[31]

The case conferences demonstrate that the assessment and communication strategies most useful for the critically ill older adult patient often require moving beyond the first-line approaches commonly used by ICU nurses. Training and modeling of these techniques are important, because communication needs and abilities of critically ill older adults are complex. Moreover, typical critical care nursing orientation programs do not include elementary or advanced approaches to communication difficulties. The case conferences presented in this article model the expert use of multiple communication modalities in a single patient interaction.

Nurses learn to accommodate common communication difficulties of critically ill older adults during these communication conferences and to plan communication strategies for future critically ill patients. For instance, adaptations to communication materials and assistive devices, such as extra large print communication boards and a hearing amplification device, are needed to accommodate hearing and vision deficits. Communication boards require active partnerships between nurses and patients to be most effective. Nurses (or other communication partners) may need to point to picture message or alphabet selections on communication boards while seeking confirmation (yes-no signal) from the patient. In these case presentations, visual "augmented input" strategies such as written key words or pointing to pictures are combined with yes-no signals or mouthing words to optimize patient attention and understanding.

Mr. Patterson's case illustrates the importance of considering and accommodating weakness and fatigue (both motor and cognitive) when establishing a consistent yes-no signal with critically ill older adults. Mr. Patterson's case also demonstrates the benefit of having a formalized avenue for transitional care regarding the patient's communication needs. Optimally, rehabilitation centers will have an opportunity to build on the communication strategies successfully used in the ICU setting. Mrs. O'Connor's case shows the value of involving families in augmentative and assistive communication strategies and reinforces the importance of adaptations for sensory impairments. To accommodate the patient's poor vision, the SLP adapted the first-letter spelling strategy to improve interpretation of mouthed words by instructing the patient to mouth the first letter of the word (rather than point to the first letter on an alphabet board).

Our discussion would be incomplete without acknowledging the facilitators and barriers that we have experienced thus far in implementing Interdisciplinary Communication Rounds in the ICU. The commitment of the SLP Department in allocating the SLP time for this activity each week is, of course, critical. The involvement of clinical leaders who serve as communication resource nurses (CRNs) is essential in generating the participation of ICU staff nurses. We seek input from CRNs and staff nurses in selecting the patient case for discussion and review. The staff members are most receptive to Communication Rounds when they need immediate help with a challenging case (e.g., Mrs. O'Connor).

Communication Rounds are a practice change in the ICU where nurses are accustomed to participating in medical rounds only on their patients and unaccustomed to consulting the SLP except for tracheostomy speaking valves or swallowing evaluations. Attendance and participation by nurses at Communication Rounds fluctuates with the activity and patient care demands in the ICU. Moreover, ICU nurses are unable to leave their patient's bedside for more than 10-15 minutes at a time to attend "teaching rounds" in another patient's room. With these constraints in mind, attendance by the patient's nurse and 1 or 2 others is considered a success. We accommodate those who cannot attend the full (approximately 20-minute) session by spreading the word to other nurses about what worked with the patient. In addition, the SLP posts the Communication Care Plan (Figure 1) at the patient's bedside, which serves as another vehicle for diffusion of information discussed at the case conference. Determining whether nurses use the AAC strategies recommended in the communication care plan and identifying which AAC strategies are most commonly used by ICU nurses after the Communication Rounds warrants further investigation.

When the SLP and CRN cannot identify a patient case to address in Communication Rounds, an alternative "teaching" format is used to maintain consistency of the rounds. In these instances, the SLP provides education on a specific topic, such as the use of hearing aids in the ICU and how to obtain and replace hearing aid batteries. Practical information of this type has been well received by the staff. This format provides discussion between individual nurses and the SLP and is often successful in reaching all nurses working on the unit that day.

Interdisciplinary Communication Rounds with SLP serve several purposes. First, the rounds reinforce new knowledge and skills in communication assessment and AAC strategies for nurses by observing a clinical expert modeling and applying those skills with real patients on their unit. In the same way, Communication Rounds expand the nurses' repertoire of AAC strategies for individual patients. Nurses are coached to consider special conditions of aging and critical illness, such as sensory (vision and hearing) impairment, motor slowing, muscle weakness, and delirium. This format moves practice change implementation directly to the bedside and produces success stories critical to changing attitudes and practice. For example, nurses who participated in Communication Rounds for Mrs. O'Connor were motivated to complete the online course later that day. These rounds begin to change nursing staff members' view of the SLP as resource for communication during the acute phase of an older adult's critical illness. Because the SLP often engages the patient in dialogue about topics outside the ICU (e.g., sports, their occupation, favorite music, or family), nurses' perceptions of the nonspeaking ICU patients as individual persons are enhanced. On several occasions, the Communication Rounds directly included the patient's family or offered suggestions to the nurse regarding how to involve the family in AAC strategies. Finally, the Communications Rounds are a model of interdisciplinary problem solving and collaboration. They provide a mechanism to identify patients who may be good candidates for more extensive SLP consultation and follow-up.

Communication case conferences shared between the speech language pathologist and the beside nurse serve to increase collaboration between speech language pathology and ICU nurses, for the benefit of both older adult patients with complex communication needs and their families. The conferences serve as a prime forum for modeling the assessment skills, communication strategies, and ACC materials required for nurses to address critically ill older adults' communication needs most adequately. With both the speech pathologist and the bedside nurse modeling and illustrating proper communicative behaviors and strategies in the ICU, these practices can spread to patients, their families, and other health care workers to improve communication with ICU patients who are unable to speak.

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